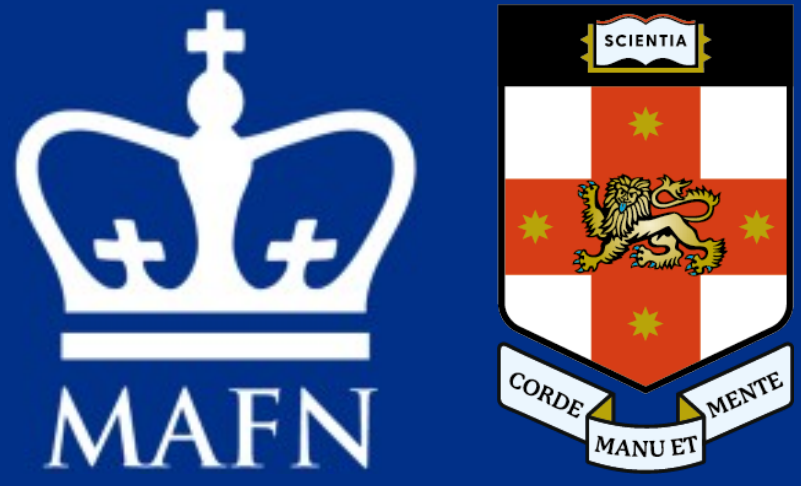




Making Tail Constraints Bind: Two Silent Failure Modes in CVaR-Constrained Portfolio RL

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Abstract

Reinforcement learning (RL) portfolio allocators are often given a tail-risk limit through a Lagrangian penalty on Conditional Value-at-Risk (CVaR). We show two ways in which this limit can fail with no visible symptom. When the penalty and the reward are on different scales, the penalty weight λ never moves away from zero, yet tail metrics still improve because of other parts of the pipeline. And the choice of test decides how the learner ranks against classical optimisers. We add a filter that checks each portfolio before it trades and moves it toward minimum variance when its forecast CVaR exceeds the budget. Out of sample, the share of weeks over budget fell from **75.1% to 7.4%**. Across 40 walk-forward folds on two ETF universes, the constraint lowered CVaR₉₉ in **26 of the 27** folds where it was active ($p = 8.8 \times 10^{-6}$).

1 Motivation

In the week of 16 March 2020 the S&P 500 fell about 15% and the VIX closed at 82.69. An RL allocator trained only for return kept its equities. The same allocator trained with a CVaR budget had already moved into Treasuries and cash, and its drawdown was 7.7% against 16.3% (Fig. 1).

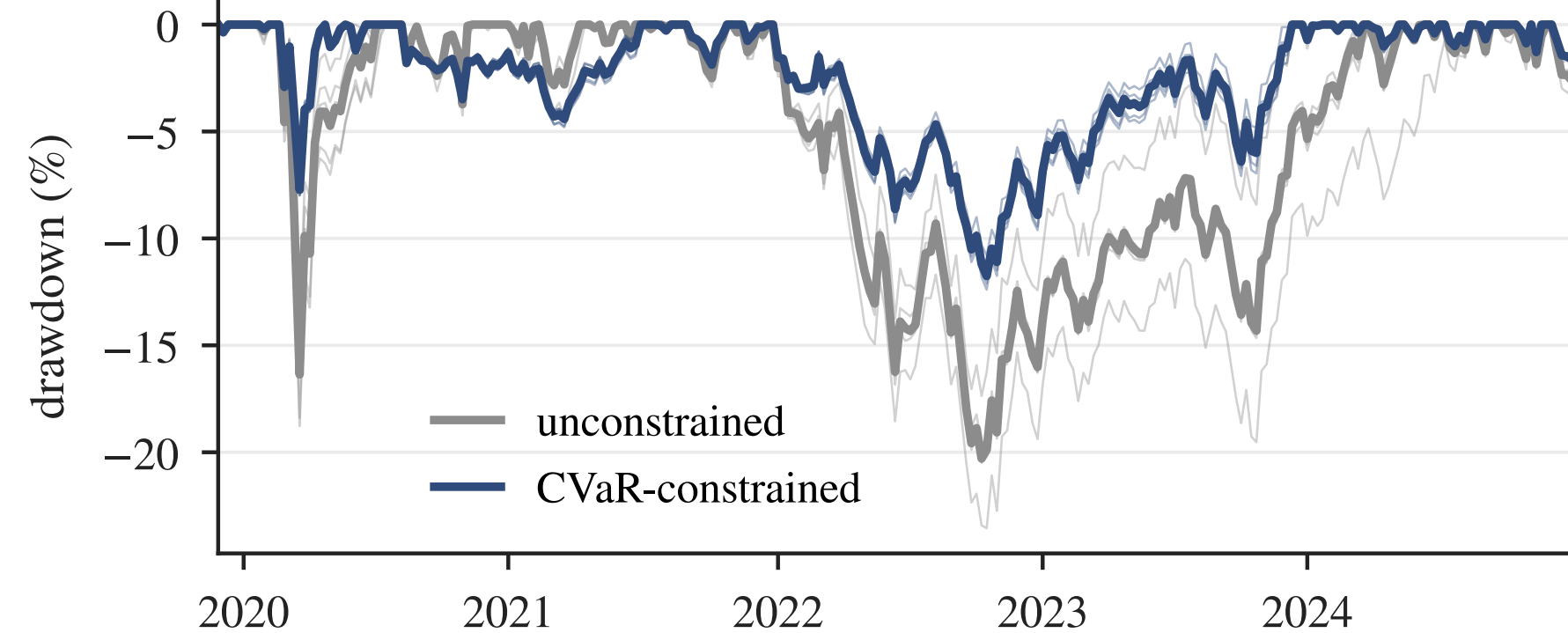


Figure 1. Drawdown over 2020 to 2024. Thick lines are five-seed means and thin lines individual seeds.

A budget of this kind is usually imposed with a Lagrangian penalty whose weight λ grows while the portfolio is over budget [2]. Section 4 asks whether λ actually grows, and Section 5 asks whether the test used to judge the result can be trusted.

2 Problem and measures

CVaR _{α} is the average loss in the worst $(1 - \alpha)$ share of weeks [1]. With $\alpha = 0.95$ it averages the worst 5%, so it measures how large the losses are beyond the Value-at-Risk (VaR) threshold (Fig. 2).

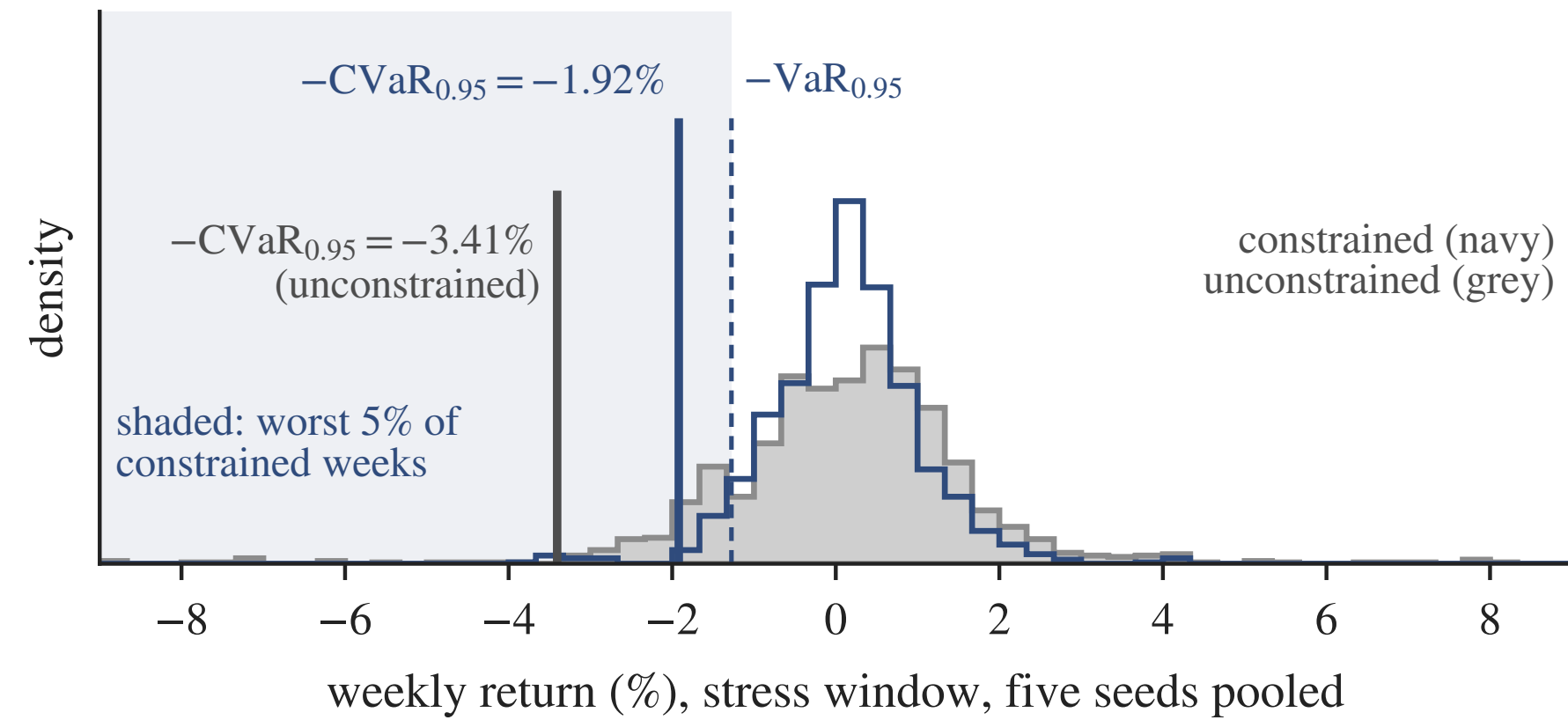


Figure 2. Weekly returns on the 2020 to 2024 stress window, five seeds pooled. The shaded left tail is the worst 5% of constrained weeks, and each line sits at minus that allocator's CVaR (per-seed means 1.89% and 3.33%).

Each week the allocator chooses long-only weights w_t on seven ETFs (SPY, TLT, HYG, DBC, GLD, UUP, BIL), using 887 weekly returns from 2008 to 2024. The problem is a constrained Markov decision process [2],

$$\max_{\pi} J(\pi) \quad \text{s.t.} \quad \text{CVaR}_{0.95}(L) \leq d = 1.2\%, \quad \text{CVaR}_{\alpha}(L) = \min_{\zeta \in \mathbb{R}} \left\{ \zeta + \frac{1}{1-\alpha} \mathbb{E}[(L - \zeta)^+] \right\}.$$

All measures use weekly net returns $r_1, \dots, r_T$ and wealth $W_t = \prod_{u \leq t} (1 + r_u)$,

$$\text{Sharpe} = \sqrt{52} \bar{r} / s_r,$$

$$\text{MDD} = \max_t (1 - W_t / \max_{u \leq t} W_u),$$

$$\widehat{\text{CVaR}}_{\alpha}(r) = -\text{mean}\{r_t : r_t \leq \hat{q}_{1-\alpha}(r)\}, \quad B = \frac{1}{|\mathcal{J}|} \sum_{t \in \mathcal{J}} \mathbf{1}[\widehat{\text{CVaR}}_{0.95}(r_{t-52:t-1}) > d],$$

where $\hat{q}_{1-\alpha}$ is the empirical quantile and B is the breach rate over the test weeks $\mathcal{J}$, of which 5% may breach. Sharpe significance uses the Probabilistic and Deflated Sharpe ratios [4], with per-week $\widehat{SR}$, skewness $\hat{\gamma}_3$, kurtosis $\hat{\gamma}_4$ and N configurations tried,

$$\text{PSR}(SR^*) = \Phi \left(\frac{(\widehat{SR} - SR^*)\sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3 \widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4} \widehat{SR}^2}} \right), \quad SR_{\text{DSR}}^* = \hat{\sigma}_{SR} \left[(1 - \gamma) \Phi^{-1} \left(1 - \frac{1}{N} \right) + \gamma \Phi^{-1} \left(1 - \frac{1}{N_e} \right) \right],$$

where γ is the Euler-Mascheroni constant.

3 Method

The allocator acts once a week. Figure 3 writes the decision as a chain of maps, and the four steps below define each one.

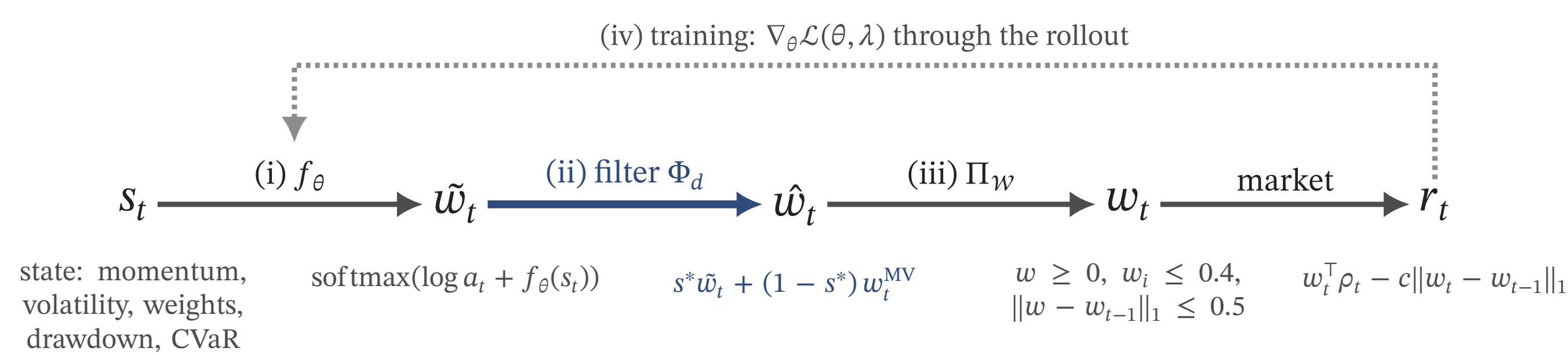


Figure 3. The decision at week t as a composition of maps.

- (i) *Policy*. An MLP f_{θ} tilts an adaptive inverse-volatility anchor a_t , $\tilde{w}_t = \text{softmax}(\log a_t + f_{\theta}(s_t))$. At initialisation $f_{\theta} \approx 0$, so training starts from the anchor and learns a state-dependent tilt.
- (ii) *Filter*. From the trailing 104 weekly returns it forecasts $\widehat{\text{CVaR}}(w) = c_{\alpha} \sigma_p(w) - \mu_p(w)$, with $c_{\alpha} = \varphi(\Phi^{-1}(\alpha))/(1 - \alpha)$ and $c_{0.95} = 2.06$. An over-budget proposal is mixed with the trailing minimum-variance portfolio w_t^{MV} , keeping as much of the proposal as the budget allows, $s^* = \max\{s \in [0, 1] : \widehat{\text{CVaR}}(s\tilde{w}_t + (1-s)w_t^{\text{MV}}) \leq d\}$. It is a safety layer in the sense of [3] and uses data up to week $t - 1$ only.
- (iii) *Projection*. Π_W enforces long-only weights, $w_t \leq 0.40$ and $\|w_t - w_{t-1}\|_1 \leq 0.50$. The market then pays r_t net of a 5 bps cost per unit of turnover.
- (iv) *Training*. On random windows of past weeks we back-propagate

$$\mathcal{L}(\theta, \lambda) = -\text{Sharpe}(r) + \lambda \text{ReLU}(\widehat{\text{CVaR}} - d) + \kappa \text{turnover}, \quad \lambda \leftarrow \max\{0, \lambda + \eta_{\lambda}(\widehat{\text{CVaR}} - d)\},$$

where $\widehat{\text{CVaR}}$ is the mean of the worst 5% of window returns, so its gradient reaches only tail weeks. The kept checkpoint has the best validation objective among those within the validation CVaR limit.

Table 1. Training and evaluation details.

component	setting
budget	$\alpha = 0.95$, $d = 1.2\%$ weekly CVaR, up to 5% of test weeks may breach
dual step	$\eta_{\lambda} = 0.001$ (mis-scaled) or 5.0 (scaled)
training	1500 updates per ablation arm and 2500 in the stress study, seeds 7, 13, 23, 42, 2025
stress test	train 2008 to 2018, test the 267 weeks from late 2019 to 2024
walk-forward	train 312, validate 52, test 26 weeks, step 26, 3-seed ensemble per fold
classical baselines	equal weight, inverse volatility, minimum variance, mean-variance, risk parity, min-CVaR LP, refit weekly on the trailing 104 weeks (52 for inverse volatility)
RL baselines	A2C, PPO (36 configurations) and SAC (8), selected on validation Sharpe
statistics	block bootstrap (2000 draws, block 4) with BCa intervals, Wilcoxon, Benjamini-Hochberg, PSR and DSR

4 Failure 1. The penalty weight never moves

With a dual step $\eta_{\lambda} = 10^{-3}$, λ stayed near zero for all 1500 training updates (Fig. 4). The CVaR excess is about 10^{-2} per week, so each update moved λ by about 10^{-5} . With $\eta_{\lambda} = 5.0$, λ rose to between 0.1 and 0.3. A model-free actor-critic failed in the same way when its cost signal (about 10^{-3}) was left unstandardised next to a unit-scale reward signal.

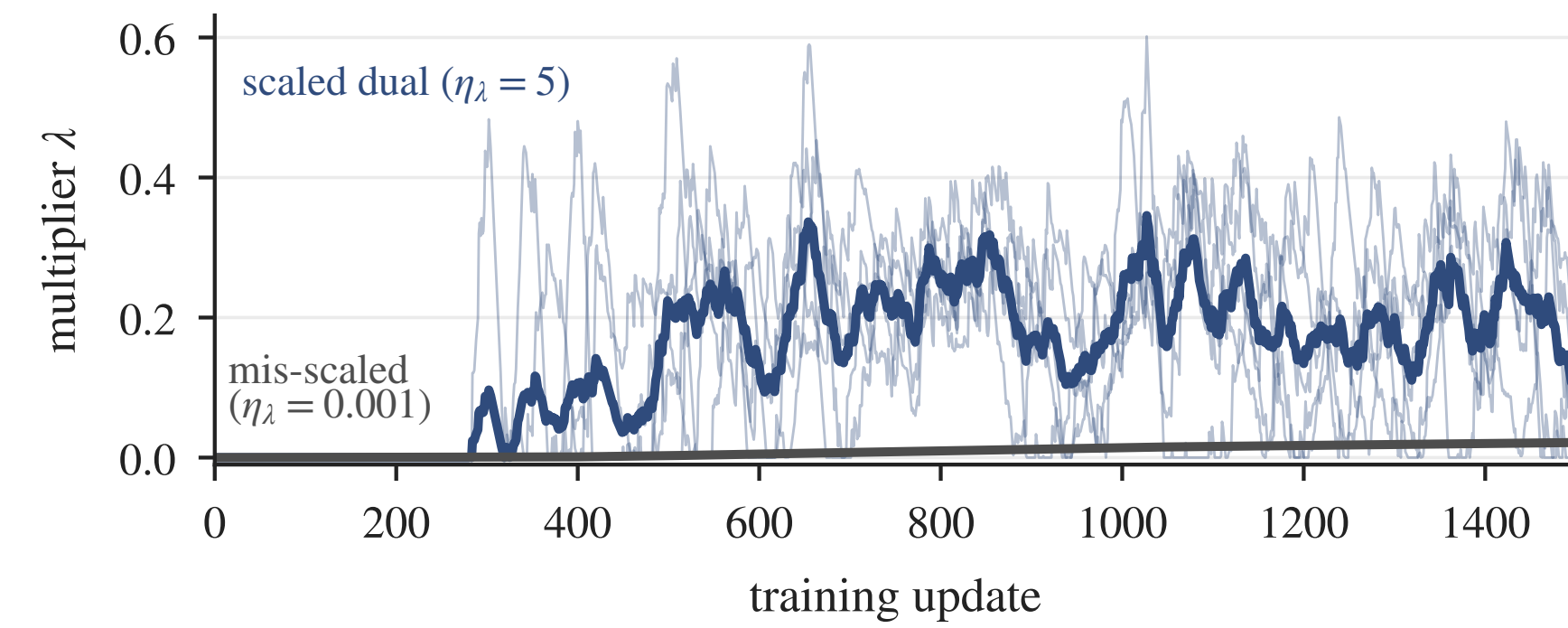


Figure 4. The penalty weight λ during training, five seeds.

Tail metrics do not reveal the failure. Even with λ near zero, the rest of the pipeline (the anchor, the projection and checkpoint selection under the CVaR limit) lowered CVaR₉₉ by 61% against the unconstrained learner (0.025 against 0.065, permutation test $p = 0.0044$).

Plot λ during training. A multiplier that stays at zero means the constraint is inactive.

5 Failure 2. The test decides the ranking

A walk-forward test retrains the model on a rolling window and evaluates it on the next unseen block. On a 31-asset universe, one fixed stress split ranked the constrained learner above minimum variance (Sharpe 0.85 against 0.23). A 20-fold walk-forward reversed the order (0.74 against 1.40, Fig. 5). Ledoit-Wolf shrinkage [5] did not change the split result (0.25). The tail-risk reduction against unconstrained RL held under both tests.

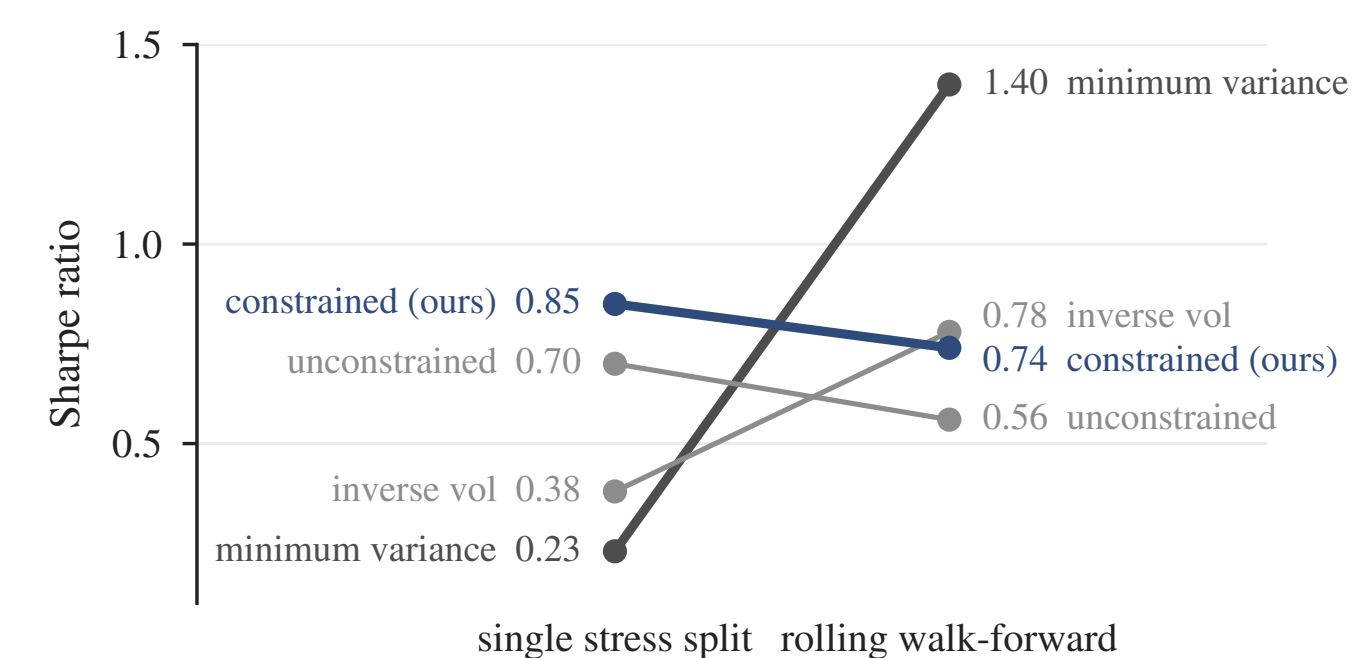


Figure 5. The same four methods ranked by two tests, 31-asset universe.

Compare learners with optimisers on rolling walk-forward folds and report the Deflated Sharpe ratio [4].

6 The filter keeps the portfolio within budget

A larger dual step made λ move, but the portfolio was then over budget in 75.1% of test weeks. Classical rules met the same budget in 92% of weeks, so the budget was attainable. Adding the filter lowered the breach rate to **7.4%** (Table 2). Raising λ during testing without the filter left it at 81.9%, so the improvement comes from the filter.

Table 2. Stress window, five-seed means. Breach is the share of test weeks over budget, with 5% tolerated.

variant	Sharpe	CVaR _{0.99}	breach
no constraint	0.626	6.5%	100.0%
penalty, $\eta_{\lambda} = 0.001$	0.987	2.5%	7.9%
penalty, $\eta_{\lambda} = 5.0$	0.911	2.8%	75.1%
penalty, λ raised during testing	0.826	3.1%	81.9%
penalty, $\eta_{\lambda} = 5.0$, with filter	0.851	1.6%	7.4%
minimum variance	0.901	4.4%	7.9%
equal weight	0.878		62.8%
risk parity	0.677		75.3%

7 Results across tests

The walk-forward test retrains every 26 weeks on the previous six years and tests on the next 26 unseen weeks, giving 20 folds per universe from 2015 to 2024. In Figure 6 each fold is one point. Below the diagonal the constraint lowered CVaR_{0.99}. On it the budget never bound, so both models held the same portfolio.

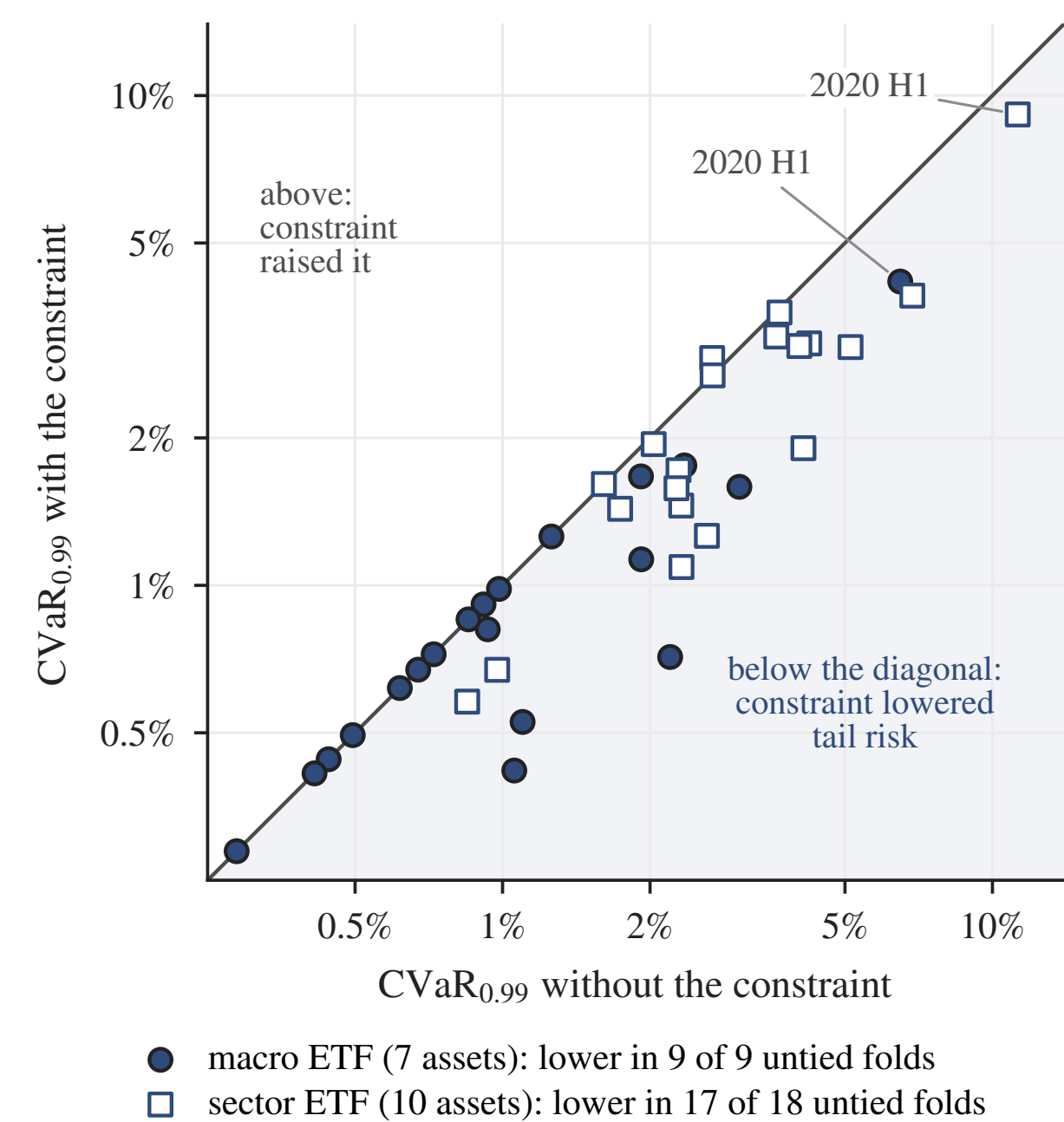


Figure 6. Walk-forward folds, CVaR_{0.99} without against with the constraint (log scale). The largest reductions came in the first half of 2020.

Table 3. Tail risk with and without the constraint in every test setting.

setting	test periods	CVaR _{0.99} without $\rightarrow$ with	evidence
stress window, 7 ETFs	267 weeks, 5 seeds	6.47% $\rightarrow$ 3.27%	95% BCa CI $[-3.5, -2.2]$ pp
stress window, 10 sector ETFs	267 weeks, 3 seeds	10.9% $\rightarrow$ 7.6%	95% BCa CI $[-5.0, -3.2]$ pp
walk-forward, 7 ETFs	20 folds	1.43% $\rightarrow$ 1.02%	lower in 9 of 9, $p = 0.0077$
walk-forward, 10 sector ETFs	20 folds	3.36% $\rightarrow$ 2.50%	lower in 17 of 18, $p = 0.0003$
both universes pooled	40 folds		lower in 26 of 27, $p = 8.8 \times 10^{-6}$
walk-forward, 31 assets	20 folds		$p = 6 \times 10^{-4}$

Fold counts exclude tied folds. Each p is a Wilcoxon signed-rank test on $\Delta_k = \text{CVaR}_{0.99,k}^{\text{con}} - \text{CVaR}_{0.99,k}^{\text{unc}}$ with $W^+ = \sum_{k: \Delta_k \neq 0} R_k \mathbf{1}[\Delta_k > 0]$ and R_k the rank of $|\Delta_k|$. Both universes pass Benjamini-Hochberg [6]. Calendar-clustered $p = 3.3 \times 10^{-4}$. PSR(0) ≥ 0.99 in both, but the Sharpe edge over classical baselines did not survive DSR.

8 Conclusion

The constraint lowered tail risk in a way that survived multiple-testing correction, and the filter achieved the out-of-sample compliance that rescaling the dual did not. The learner did not beat rolling minimum variance on Sharpe out of sample. The study uses weekly data with proportional costs, and the Gaussian forecast can understate fat tails.

References. [1] R. T. Rockafellar, S. Uryasev. Optimization of conditional value-at-risk. *J. Risk* 2 (2000). [2] E. Altman. *Constrained Markov Decision Processes*. 1999. [3] G. Dalal et al. Safe exploration in continuous action spaces. arXiv:1801.08757, 2018. [4] D. H. Bailey, M. López de Prado. The deflated Sharpe ratio. *JPM* 40 (2014). [5] O. Ledoit, M. Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *JMVA* 88 (2004). [6] Y. Benjamini, Y. Hochberg. Controlling the false discovery rate. *JRSS B* 57 (1995).
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 GitHub.com/nl2992/ICAIF_cvar-rl-portfolio-allocator

